

We collect CRS reports that were published before May 20, 2020 from EveryCRSReport⁷ where the original PDF files are already parsed into HTML. We only keep documents with expert-written summaries. We then gather texts from the html files.

B Experiment Details

FactCC Training Data Construction. Kryscinski et al. (2020) generate training data by applying rule-based transformations to sentences from source documents. We leverage reference summaries, where we train a FactCC model by reading a summary sentence (i.e., the claim) and a context to predict the corresponding label. A context is constructed by greedily selecting sentences that maximize the improvement of its ROUGE-2 when compared against the reference summary sentence. Following FactCC, we apply *sentence negation*, *entity swap*, and *number swap* to summary sentences to construct negative claims and use the original sentences as positive claims. During testing, we first find the context for each system summary sentence. The model then predicts a sentence-level faithfulness score by reading the system summary sentence and the context.

Evaluation Model Training. We fine-tune BERT (Devlin et al., 2019) for both FactCC and QA models. We include an additional classification head to predict entailment label or answer spans based on the [CLS] token. For GovReport dataset, we consider a base version of BERT with uncased tokens. For PubMed, we use a BERT model which is fine-tuned on PubMed abstracts to obtain better performance⁸.

Entity Extraction Model. We use OneIE to extract entities from the reference summary (Lin et al., 2020). OneIE is a unified framework that combines entities, relations, and events extraction in one model. The model leverages the BERT pre-trained weights as the sentence embedding to produce entities, relations, and events from a sentence. Two OneIE models are built.

The first model for government reports is trained on the Automatic Content Extraction (ACE) 2005 dataset (Walker et al., 2006). This model can extract entities from general conversation contexts

⁷<https://www.everycrsreport.com>

⁸https://huggingface.co/monologg/biobert_v1.0_pubmed_pmc

	Genia 2011	Genia 2013	PubMed
Entity Type			
Anaphora	-	105	-
Entity	480	121	-
CellLine	-	-	614
Chemical	-	-	14,051
Disease	-	-	62,228
Mutation	-	-	164
Protein	11,539	3,562	15,577
Species	-	-	52,954
Event Type			
Binding	880	167	-
Gene Expression	2,076	666	-
Localization	264	44	-
Negative Regulation	338	273	-
Phosphorylation	175	105	-
Positive Regulation	1,123	311	-
Protein Catabolism	100	23	-
Protein Modification	-	8	-
Regulation	292	72	-
Transcription	580	97	-
Ubiquitination	-	4	-

Table 9: Dataset description for training OneIE Biomedical extraction. While Genia 2011 and 2013 datasets focus more on event extraction, PubMed covers more entities.

such as *People*, *Location*, or *Organization*, and events such as *Movement*, *Conflict*, or *Justice*, etc.

The second model for scientific domain information extraction is trained on the Genia 2011 (BioNLP, 2011), Genia 2013 (BioNLP, 2013), and PubMed (Wei et al., 2019) datasets. It extracts entity such as *Gene*, *Variant*, *Disease*, *Chemical*, or *Species*, and events such as *Gene Expression*, *Binding*, *Protein Modification*, or *Positive Regulation*, etc. The full list of entity and event types can be found in Table 9. To train this model, we fine-tune the BioBERT pre-trained model (Lee et al., 2020) on the COVID-19 Open Research (CORD-19) dataset (Wang et al., 2020b). As we proposed, this model is applied to the PubMed data.

C Additional Sample Outputs

We include two samples from GovReport and PubMed to further illustrate that our model with HEPOS attention generates more faithful and informative summaries in Fig. 7 and Fig. 8.

D Human Evaluation Guideline

In human evaluation, annotators are asked to evaluate the system summaries generated for a report or a paper. In addition to the summaries, annotators are provided with *the report or the paper to be summarized* and a corresponding *human-written*

reference. Human judges evaluate each system summary **sentence by sentence**. The annotation consists of three tasks, which are described below.

Task 1: Aspect Labeling. First, annotators are asked to decide which **aspect** each sentence belongs to. For government reports, each sentence should be categorized into three aspects: (1) Why GAO did this study, (2) What GAO found, and (3) What GAO recommends. For scientific papers, summaries have four aspects: (1) Introduction and Literature, (2) Methods, (3) Results, and (4) Discussion and Conclusion. Table 10 and Table 11 contain example reference summaries with labeled aspects.

Task 2: Sentence-level Faithfulness Error Labeling. Next, annotators will judge whether each sentence contains any **unfaithful content**. Unfaithful content is categorized into three types. A “0” or “1” label will be given to each type, where “0” indicates the sentence is free of such type of error, and “1” otherwise.

Concretely, unfaithful content is the fabricated or contradictory content which *is not present or contradicts the facts* in the source article. It can also be *ambiguous expression* which distorts the meaning. Here are detailed descriptions for the three types of errors:

- **Hallucination** error refers to fabricated content that cannot be found or inferred from the source.
- Misconstruction error that is due to **deletion** of entities, events, or clauses, resulting in sentences that are incomplete, missing context, or ungrammatical.
- Misconstruction error that is caused by **false concatenation** of content from different places in the source.

Task 3: Aspect-level Summary Quality Rating. After reading the full summary, annotators will evaluate the **informativeness** of the summary for each aspect— whether the summary provides *a necessary and enough coverage of information* in the *reference*. For instance, whether the summary covers all the salient points in “Why GAO did this study”.

Here are detailed descriptions of informativeness:

- **5:** Summary covers enough key points in the

reference (only misses minor topics), and is free of unfaithful errors.

- **4:** Summary covers major key points (e.g., 80 percent) and may miss one or two key points in the reference. Summary can contain one unfaithful error.
- **3:** Summary covers roughly half of the key points in the reference or contains 2 or 3 unfaithful errors.
- **2:** Summary only covers 1 or 2 key points and misses many important topics (e.g. > 80 percent) in the reference, or contains more than 3 major unfaithful errors, e.g. summary fabricates or distorts some facts.
- **1:** Summary is irrelevant and does not cover any content in the reference.

Summary by Human:*What GAO found*

The Food and Drug Administration (FDA) and U.S. Department of Agriculture (USDA) have begun collaborating on regulatory oversight of cell-cultured meat. For example, in 2019, the agencies signed an interagency agreement and created three working groups to carry out the terms of the agreement. However, the agreement and working groups could more fully incorporate practices to enhance and sustain collaboration, such as defining outcomes. For example, the agreement identifies the development of labeling principles as an outcome, but does not describe how the agencies will track and monitor progress toward this outcome, and the working groups identify a lead agency but not members' roles. (...)

Developing and updating written guidance and agreements is also a leading practice for interagency collaboration. By fully incorporating leading practices into their efforts to collaborate, the agencies could minimize potential overlap and fragmentation, use resources in a more efficient manner, and better ensure the public and other key stakeholders have clarity about the agencies' oversight responsibilities. (...)

Model w/ full attn.:*What GAO found*

The frequency of inspections the agencies conduct varies, as follows: FDA. FDA conducts risk-based inspections of high- risk and non-high-risk food facilities. FDA's authority requires a risk- based approach, in which inspection rates vary depending on the level of risk associated with a food product. (...)

In 2010, FDA began conducting inspections of meat slaughter facilities to determine whether they were complying with the agency's longstanding interpretation of its statutes requiring it to do so. In addition, USDA conducted inspections of slaughter facilities that process catfish to determine if they are complying with USDA's requirement to inspect catfish slaughter facilities. USDA also conducts inspections of processing facilities to verify that they are following USDA's requirements to inspect these facilities. In 2011, USDA began inspecting processing facilities for catfish for the first time. (...)

Model w/ HEPOS enc-dec attn.:*What GAO found*

The Departments of Health and Human Services (HHS) and Agriculture have taken several steps to coordinate their oversight of the development, commercialization, and regulation of cell cultured meat and seafood, but they have not fully incorporated leading practices for interagency collaboration. For example, HHS and USDA have established interagency working groups that have met to discuss issues such as coordination and coordination. However, the agencies have not yet fully incorporated some leading practices, such as establishing clear roles and responsibilities, into their interagency agreements. (...)

Figure 7: Sample summaries for a government report. Model with truncated input generates **unfaithful content**. Our HEPOS encoder-decoder attention with Sinkhorn encoder attention covers more **salient information** in "What GAO found" aspect.